Closing the Gap East of the River

Healthy Economies, Healthy Communities

Team Data Benders | Washington, DC — Wards 7 & 8 | 2026 Mastercard Data Challenge

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2 Objective & Problem Statement

PROBLEM STATEMENT

Poor economic conditions in East of the River communities reduce healthcare access indirectly — through food affordability, transportation, commute time to work, poor internet access, and low neighborhood service availability. Small businesses can help close these gaps faster than large system reforms.

Why East of the River?

IGS < 45	All 46 EOTR tracts score below 45, with a median of 38, 18 points below the rest of DC (56). The Economy pillar is the weakest at just 33.
100% MUA	Every analyzed tract is within a federally designated Medically Underserved Area (HRSA), confirming broad healthcare access disadvantage.
Structural	Economy has been EOTR's lowest pillar every year from 2017 to 2025. This is persistent, not cyclical and is driven by decades of disinvestment.

Guiding Research Questions

1. How do economic conditions in EOTR translate into measurable healthcare access barriers?

2. Which specific IGS indicators (Economy, Place, Community) are most predictive of health burden?

3. Can small-business interventions — clinics, food retail, digital hubs — address the identified barriers?

Sources: Mastercard IGS, HRSA MUA Designations, ACS, CDC BRFSS PLACES, HUD, USDA, SBA

3 Community Overview

Washington, DC is one of America's most economically vibrant cities — but the Anacostia River divides it into two starkly different realities. East of the River, Wards 7 & 8 encompass 46 census tracts where persistent economic weakness, concentrated poverty, and structural barriers have compounded over decades of underinvestment.

Demographics

Population: 150,000 residents across 46 tracts. Predominantly Black communities. Median income significantly below DC average.

Economic Conditions

Economy pillar score: 33 vs 56 (rest of DC). 40% poverty rate in weakest tracts. Labor market engagement among lowest in DC.

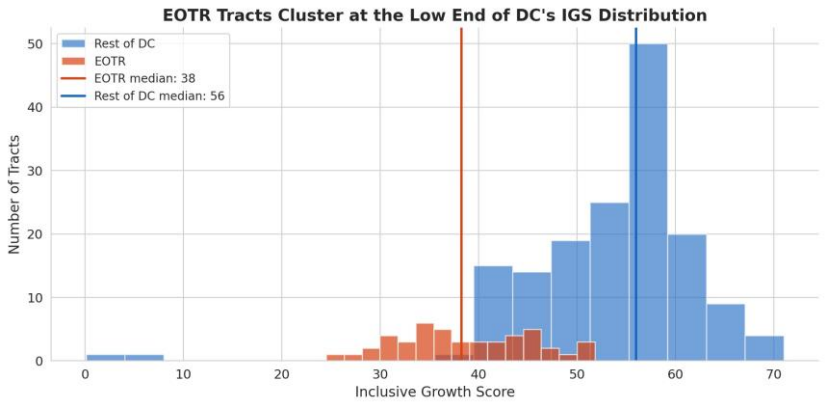
Social Context

Historically redlined neighborhoods. Limited commercial diversity. High transit dependence — 2× longer commutes on public transit vs car.

Sources: Mastercard IGS 2017–2025 (averaged), ACS 5-Year, CDC BRFSS PLACES 2023

EOTR Tracts Cluster at the Low End of DC's IGS Distribution

EOTR's median IGS of 38 falls 18 points below the rest of DC's median (56). The histogram shows EOTR tracts concentrated at the bottom of the distribution — a clear structural divide, not a statistical outlier.



4 IGS Benchmarking

EOTR trails the rest of DC on every IGS dimension. The gap is statistically confirmed (Mann-Whitney U, $p < 0.001$) and has persisted across all available years (2017–2025). Using the State benchmark, EOTR's median IGS of 38 places it 18 points below the rest of DC's median of 56.

IGS Pillar	EOTR Median	Rest of DC	Gap	Statistical Sig.
Overall IGS	38	56	**−18**	$p < 0.001$ (Mann-Whitney)
Economy	33	56	**−23**	$p < 0.001$ — weakest every year
Place	42	55	**−13**	$p < 0.001$
Community	40	54	**−14**	$p < 0.001$

Weakest Sub-Indicators Within EOTR

Economy
Labor Market Engagement
Commercial Diversity Score

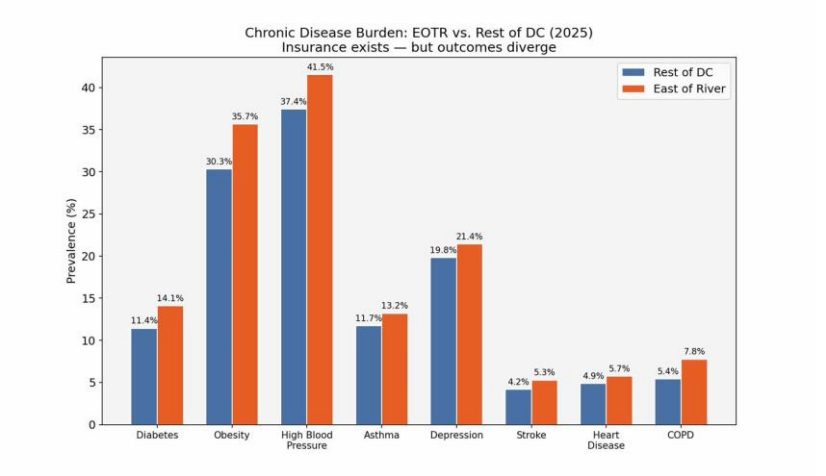
Community
Female Above Poverty
Health Insurance Coverage

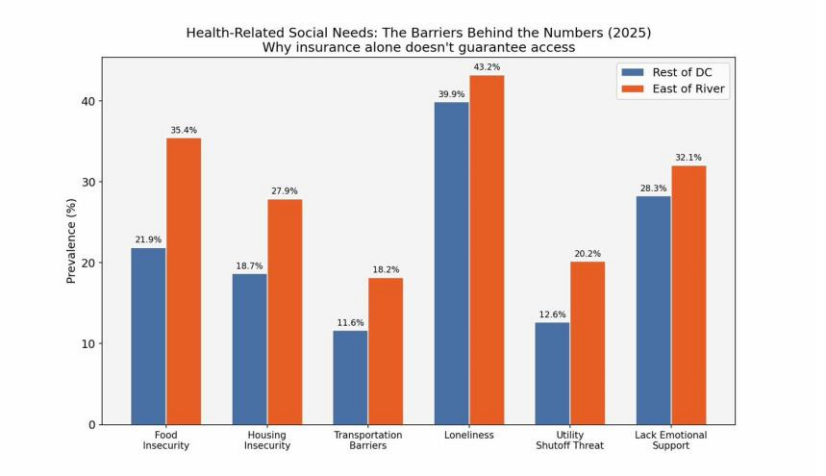
Place
Internet Access Score
Travel Time to Work

Source: Mastercard IGS, State (DC) benchmark, 2017–2025 averaged scores

5 Key Findings

Across 8+ independent datasets, the evidence converges: economic weakness in EOTR translates directly into measurable health burden. Machine learning confirms that business metrics (Commercial Diversity, Small Business Loans) carry independent predictive power for health outcomes.





40% vs 27%

Food insecurity in bottom 10 vs top 10 economy tracts ($p = 0.014$)

21% vs 14%

Transportation barriers in bottom 10 vs top 10 economy tracts ($p = 0.017$)

$\rho > 0.8$

Labor market ↔ HUD index correlation validates IGS

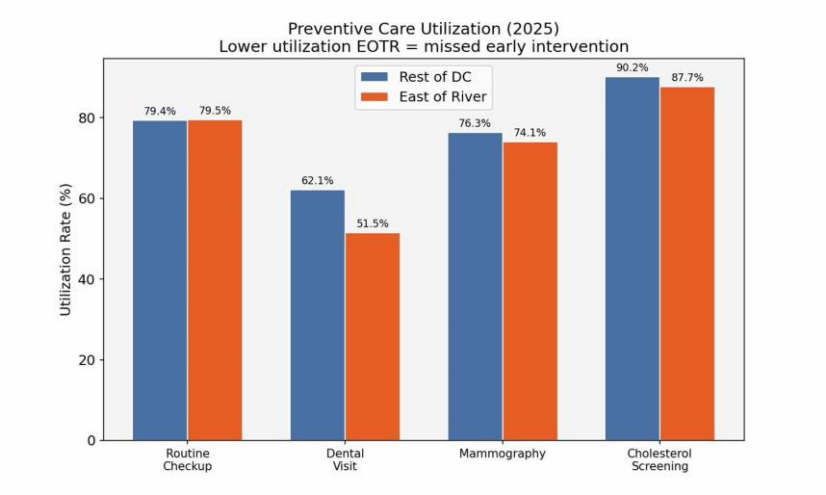
Lasso

Commercial Diversity + SB Loans survive regularization

Sources: Mastercard IGS, CDC BRFSS PLACES 2023, HUD Labor Market Index, ACS.

6 Healthcare Access Context

Being near a clinic doesn't help if you can't afford to walk in. Insurance exists — but outcomes diverge.



Five Barrier Mechanisms

- **Food Affordability**
  ~40% poverty in weakest economy tracts; food and housing costs crowd out healthcare spending.
- **Transportation**
  Transit commuters face 45+ min at 2× the rate of drivers — greatest time cost for medical visits.
- **Digital Exclusion**
  Weakest labor-market tracts have substantially higher digital disconnection rates, limiting telehealth.
- **Housing Strain**
  Renter-heavy tracts face severe cost burden, reducing disposable income available for healthcare.
- **Service Gaps**
  Less commercially diverse tracts have significantly higher food insecurity. Distance ≠ access.

Key Insight: Highest food-insecurity tracts are actually closer to healthcare facilities. Barriers are economic and logistical, not geographic.

Sources: CDC BRFSS PLACES 2023, USDA Food Access Atlas, DC Built Environment Indicators, ACS Commute & Housing Tables

7 Proposed Small-Business Solutions

77% of DC businesses have <20 employees. Healthcare (+7.9%) and Food Services (+62.6%) are DC's fastest-growing sectors — directly aligned with EOTR's gaps. Our solution? One integrated Community Health Access Network — three service layers — all within SBA-qualified EOTR tracts

1 Health Layer

Micro-Clinics & Mobile Health

Neighborhood-embedded small-format clinics and mobile health units that reduce travel burden and lower cost. Directly addresses the affordability and transportation barriers confirmed in the analysis — particularly for the 40% poverty rate in weakest economy tracts.

Targets: Affordability + Transportation barriers

2 Food Layer

Affordable Food Retail & Meals

Small food businesses offering affordable, accessible options in commercially thin tracts. Analysis proved supermarket distance does not explain food insecurity — affordability does. This layer targets the 40% food insecurity rate in bottom economy tracts.

Targets: Food insecurity + Service gaps

3 Digital Layer

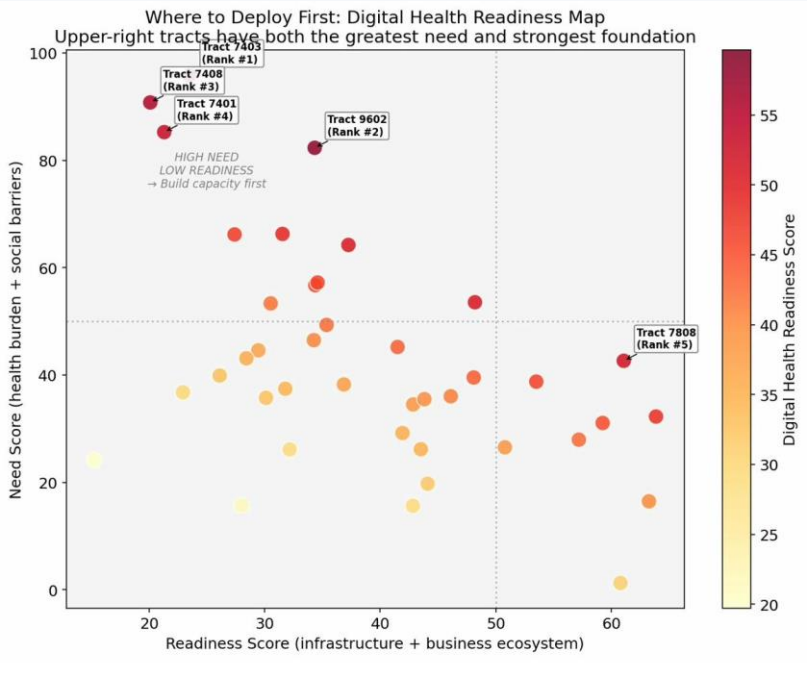
Telehealth Access Hubs

Telehealth stations and health info kiosks hosted inside existing businesses — pharmacies, barbershops, laundromats. Addresses the digital exclusion barrier by meeting residents where they already go, bypassing the need for home broadband.

Targets: Digital exclusion + Transportation

Sources: Lightcast DC Industry Data (2019–2024), SBA Census Tract Designations

8 Implementation Considerations



Deployment Strategy

Capital & Funding

96% of EOTR tracts qualify for SBA incentives. Leverage New Markets Tax Credits, CDFIs, and Mastercard inclusive growth programs. Start-up costs offset by existing federal designations.

Partnerships

Anchor institutions — hospitals, churches, universities — as distribution partners. DC's Neighborhood Prosperity Fund and existing community health workers as connectors.

Workforce Pipeline

DC Infrastructure Academy (Ward 8) already trains residents for Healthcare and Food Services — DC's two fastest-growing sectors (+7.9% and +62.6%). Business interventions plug directly into this pipeline.

Phasing & Prioritization

Start with 5 highest-need tracts identified by the readiness map (left): lowest Commercial Diversity + highest food insecurity. Expand based on measured impact using IGS tracking.

96% of EOTR tracts already qualify for federal business incentives (vs 22% rest of DC) — the infrastructure for investment exists.

Sources: SBA Census Tract Designations (July 2023), Lightcast DC Labor Data, DC Infrastructure Academy, Digital Health Readiness Model

9 Metrics of Success

All metrics align with existing IGS indicators — progress can be tracked year-over-year using the same platform that identified the problem.

Outcome Metric	Baseline	IGS Indicator	Solution Layer	Measurement
Commercial Diversity	26 / 100	Economy Pillar	All Layers	IGS annual score update
Food Insecurity Rate	34%	Community (BRFSS)	Food Layer	CDC PLACES annual estimate
Small Business Loans	42 / 100	Economy Pillar	Capital Access	IGS annual score update
Transport Barriers	17%	Place (BRFSS)	Health Layer	CDC PLACES + ACS commute
Digital Disconnection	Varies	Internet Access	Digital Layer	ACS broadband subscriptions

ML Model Validation

Lasso regression retained Commercial Diversity Score (coeff: −1.217) and Small Business Loans Score (coeff: −1.406) as independent predictors of food insecurity after controlling for labor market conditions, housing, and internet access. Decision tree analysis confirmed Small Business Loans as a meaningful decision node. These metrics should be prioritized for tracking intervention impact.

Sources: Mastercard IGS, CDC BRFSS PLACES 2023

3 grocery stores.

That's all Wards 7 and 8 have for over 150,000 residents.

5 miles west

Ward 3

Life expectancy: **86 years**

Median income: **$112,873**

East of the River

Ward 8

Life expectancy: **69 years**

Median income: **$30,910**

17 years of life. Same city. One river apart.

Our data shows that small-business investment is statistically linked to lower food insecurity, fewer transportation barriers, and reduced mental distress. 96% of EOTR tracts already qualify for federal programs.

If the evidence, the tools, and the funding pathways all exist — what are we waiting for?

DATA SOURCES & ACKNOWLEDGEMENTS

Mastercard IGS • ACS 5-Year • CDC BRFSS PLACES • HUD Labor Market Index • USDA Food Access Atlas • HRSA MUA • SBA Designations • Lightcast DC Labor Data • DC Built Environment Indicators

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